Q2 -- Name:

Question 1 A binary classifier uses the model $q(\mathbf{x}; \mathbf{w}, b) = \mathbf{w}^T\mathbf{x} + b$ with $\mathbf{w} = [1, -2]^T$ and $b = 3$.

a) Compute q for the test point $\mathbf{x}_0 = [2, 1]^T$.

b) If the decision rule is: class A when $q > 0$, class B when $q \le 0$, which class does $\mathbf{x}_0$ belong to?

Question 2 A model makes predictions on two training examples. The squared-error loss on one example is $e_i^2 = (\text{predicted}_i - \text{actual}_i)^2$, and the total loss is $L = \Sigma\, e_i^2$. Given:

i	Predicted	Actual
1	1.2	1.0
2	0.4	0.1

Compute the total loss L.

Question 3 Consider the loss function $L(w_0, w_1) = w_0^2 + 2w_1^2$.

a) Compute the gradient ∇L at the point $(w_0, w_1) = (2, -1)$.

b) Perform **one** gradient descent step with learning rate $\eta = 0.1$. What are the new weights?

Question 4 True or False (circle one and briefly explain):

T / F: The gradient of a loss function points in the direction of steepest *increase* in L. Therefore, to decrease L most rapidly, we move in the *negative* gradient direction.

Explanation:

Question 5 What is the shape of the level contours $L(w_0, w_1) = w_0^2 + 4w_1^2 = c$? (Circle one)

1. Circles centered at the origin
2. Ellipses centered at the origin
3. Straight lines through the origin
4. Parabolas opening upward

Question 6 For $f(x) = x^2$, use the first-order Taylor approximation expanded around $x = 3$ to estimate $f(3.1)$. Show the formula and your computation.